

```
pip install transformers datasets torch scikit-learn
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
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Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
```

```
from transformers import AutoTokenizer, AutoModelForSequenceClassification, Trainer, TrainingArguments
from datasets import load_dataset
import torch
```

```
dataset = load_dataset("imdb")
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
README.md:      7.81k/? [00:00<00:00, 399kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
plain_text/train-00000-of-00001.parquet: 100%                21.0M/21.0M [00:01<00:00, 101MB/s]
plain_text/test-00000-of-00001.parquet: 100%                20.5M/20.5M [00:00<00:00, 102MB/s]
plain_text/unsupervised-00000-of-00001.p(...): 100%          42.0M/42.0M [00:01<00:00, 49.6MB/s]
Generating train split: 100%                                25000/25000 [00:00<00:00, 34568.68 examples/s]
Generating test split: 100%                                25000/25000 [00:00<00:00, 53433.31 examples/s]
Generating unsupervised split: 100%                         50000/50000 [00:00<00:00, 59058.14 examples/s]

```

```
text = text.lower()
```

```

-----
NameError                                Traceback (most recent call last)
/tmp/ipykernel_217/2080091495.py in <cell line: 0>()
----> 1 text = text.lower()

NameError: name 'text' is not defined

```

```
tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
```

```

config.json: 100%                570/570 [00:00<00:00, 12.3kB/s]
tokenizer_config.json: 100%       48.0/48.0 [00:00<00:00, 557B/s]
vocab.txt:      232k/? [00:00<00:00, 2.70MB/s]
tokenizer.json:  466k/? [00:00<00:00, 3.29MB/s]

```

```

def preprocess_function(examples):
    return tokenizer(examples["text"], truncation=True)

tokenized_imdb = dataset.map(preprocess_function, batched=True)

```

```

Map: 100%                25000/25000 [00:36<00:00, 873.01 examples/s]
Map: 100%                25000/25000 [00:29<00:00, 820.95 examples/s]
Map: 100%                50000/50000 [01:05<00:00, 880.53 examples/s]

```

```

tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")

def tokenize(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)

dataset = dataset.map(tokenize, batched=True)

```

```

config.json: 100%                483/483 [00:00<00:00, 36.9kB/s]
tokenizer_config.json: 100%       48.0/48.0 [00:00<00:00, 1.08kB/s]
vocab.txt:      232k/? [00:00<00:00, 2.45MB/s]
tokenizer.json:  466k/? [00:00<00:00, 6.31MB/s]
Map: 100%                25000/25000 [00:40<00:00, 518.16 examples/s]
Map: 100%                25000/25000 [00:33<00:00, 834.28 examples/s]
Map: 100%                50000/50000 [01:08<00:00, 797.89 examples/s]

```

```

train_dataset = dataset["train"]
test_dataset = dataset["test"]

```

```

train_dataset.set_format(type="torch", columns=["input_ids", "attention_mask", "label"])
test_dataset.set_format(type="torch", columns=["input_ids", "attention_mask", "label"])

```

```
model = AutoModelForSequenceClassification.from_pretrained("distilbert-base-uncased", num_labels=2)
```

model.safetensors: 100%

268M/268M [00:02<00:00, 168MB/s]

Loading weights: 100%

100/100 [00:00<00:00, 308.04it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status	
-----+-----+		
vocab_transform.weight	UNEXPECTED	
vocab_transform.bias	UNEXPECTED	
vocab_projector.bias	UNEXPECTED	
vocab_layer_norm.weight	UNEXPECTED	
vocab_layer_norm.bias	UNEXPECTED	
classifier.weight	MISSING	
pre_classifier.weight	MISSING	
classifier.bias	MISSING	
pre_classifier.bias	MISSING	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task

```
training_args = TrainingArguments(  
    output_dir="./results",  
    eval_strategy="epoch",  
    num_train_epochs=2,  
    per_device_train_batch_size=8  
)
```

```
from transformers import Trainer  
  
trainer = Trainer(  
    model=model,  
    args=training_args,  
    train_dataset=train_dataset,  
    eval_dataset=test_dataset  
)  
  
trainer.train()
```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but no pin_memory_device is provided. It will default to cudaDeviceNone

super().__init__(loader)

[249/6250 1:58:57 < 48:10:02, 0.03 it/s, Epoch 0.08/2]

Epoch	Training Loss	Validation Loss
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Start coding or [generate](#) with AI.